

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Formative Assessment 4

Formative Assessment (FA-4) - Answer Key

Class: Class 9

Assessment Type: Formative Assessment (FA-4)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Web Technologies - HTML, CSS, JavaScript

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

1. **Answer:** b) <nav>
 - **Explanation:** The <nav> HTML5 tag is used to define a section containing navigation links.
2. **Answer:** c) display: flex
 - **Explanation:** display: flex creates a flexible layout container, enabling flexible alignment and distribution of child elements.
3. **Answer:** b) Changes the HTML content of element
 - **Explanation:** innerHTML sets or returns the HTML content (inner markup) of an element.
4. **Answer:** c) onclick
 - **Explanation:** The onclick event occurs when the user clicks on an HTML element. onmouseover triggers on hover, onchange on form element changes.
5. **Answer:** b) rem
 - **Explanation:** rem (root em) is relative to the font-size of the root <html> element. em is relative to the parent element's font-size.

Section II: Short Answer (5 x 2 = 10 marks)

1. **Differentiate between inline, internal, and external CSS.**
 - **Answer:**
 - **Inline CSS:** Applied directly to an element using the style attribute. Used for single-element styling.
 - **Internal CSS:** Defined in a <style> tag within the <head> section. Used for single-page styling.
 - **External CSS:** Defined in a separate .css file linked via <link> tag. Used for multi-page styling and better organisation.
 - **Marking:** 1 mark for any two correct descriptions with use cases, 1 mark for the third.
2. **Write HTML and CSS for a responsive navigation bar structure.**
 - **Answer:**

```
<nav class="navbar">
  <div class="logo">School</div>
  <ul class="nav-links">
    <li><a href="#">Home</a></li>
    <li><a href="#">About</a></li>
    <li><a href="#">Contact</a></li>
  </ul>
  <div class="hamburger">&#9776;</div>
</nav>
```

```

.navbar { display: flex; justify-content: space-between; align-items: center;
padding: 1rem; }
.nav-links { display: flex; list-style: none; gap: 2rem; }
.hamburger { display: none; font-size: 1.5rem; }
@media (max-width: 768px) {
    .nav-links { display: none; }
    .hamburger { display: block; }
}

```

- **Marking:** 1 mark for HTML structure with nav element, 1 mark for CSS flexbox and media query.

3. Explain the CSS box model and box-sizing: border-box.

- **Answer:** The CSS box model consists of content, padding, border, and margin (from innermost to outermost). box-sizing: border-box changes the default behaviour so that padding and border are included in the element's total width and height, rather than being added on top.
- **Marking:** 1 mark for explaining the four components, 1 mark for explaining border-box behaviour.

4. Write JavaScript to validate a form field.

- **Answer:**

```

function validateForm() {
    var field = document.getElementById("name").value;
    if (field.trim() === "") {
        alert("Error: Field cannot be empty!");
        return false;
    }
    alert("Valid! Value: " + field);
    return true;
}

```

- **Marking:** 1 mark for checking empty field, 1 mark for displaying alert with value or error message.

5. Difference between var, let, and const.

- **Answer:**
 - var: Function-scoped, can be redeclared and updated. Hoisted.
 - let: Block-scoped, can be updated but not redeclared in same scope.
 - const: Block-scoped, cannot be updated or redeclared. Must be initialized.
- **Marking:** 1 mark for any two correct descriptions with examples, 1 mark for the third.

Section III: Long Answer (5 x 1 = 5 marks)

1. Create a Student Management System web page.

- **Answer:**

```

<!DOCTYPE html>
<html>
<head>
    <style>
        body { font-family: Arial; background: #f4f4f4; padding: 20px; }
        .form-container { background: white; padding: 20px; border-radius: 8px;
max-width: 500px; margin: auto; }
        input { width: 100%; padding: 10px; margin: 8px 0; border: 1px solid #ddd;
border-radius: 4px; box-sizing: border-box; }
        button { background: #007bff; color: white; padding: 10px 20px; border:

```

```

none; border-radius: 4px; cursor: pointer; }
    .result-card { background: #e8f5e9; padding: 15px; border-radius: 8px;
margin-top: 15px; display: none; }
</style>
</head>
<body>
    <div class="form-container">
        <h2>Student Report Card</h2>
        <input type="text" id="name" placeholder="Student Name">
        <input type="text" id="class" placeholder="Class">
        <input type="text" id="section" placeholder="Section">
        <input type="number" id="roll" placeholder="Roll Number">
        <input type="number" id="sub1" placeholder="Subject 1 Marks">
        <input type="number" id="sub2" placeholder="Subject 2 Marks">
        <input type="number" id="sub3" placeholder="Subject 3 Marks">
        <button onclick="calculate()">Calculate</button>
        <div class="result-card" id="result"></div>
    </div>
    <script>
        function calculate() {
            var s1 = parseFloat(document.getElementById("sub1").value);
            var s2 = parseFloat(document.getElementById("sub2").value);
            var s3 = parseFloat(document.getElementById("sub3").value);
            if (isNaN(s1) || isNaN(s2) || isNaN(s3) || s1 < 0 || s1 > 100 || s2 < 0
|| s2 > 100 || s3 < 0 || s3 > 100) {
                alert("Please enter valid marks (0-100)");
                return;
            }
            var total = s1 + s2 + s3;
            var percentage = (total / 300 * 100).toFixed(2);
            var grade = percentage >= 90 ? "A" : percentage >= 75 ? "B" :
percentage >= 60 ? "C" : percentage >= 45 ? "D" : "F";
            var resultDiv = document.getElementById("result");
            resultDiv.style.display = "block";
            resultDiv.innerHTML = "<h3>Result</h3><p>Total: " + total + "</
p><p>Percentage: " + percentage + "%</p><p>Grade: " + grade + "</p>";
        }
    </script>
</body>
</html>

```

- **Marking:** 1 mark for HTML form with all fields, 1 mark for CSS styling, 1 mark for calculate function, 1 mark for grade logic and result display, 1 mark for input validation.

Part B: Hands-on Practical Lab Task (5 marks)

- **Answer:**

```

<!DOCTYPE html>
<html>
<head>
    <style>
        body { font-family: Arial; text-align: center; background: #1a1a2e; color:
white; }
        .quiz-container { max-width: 600px; margin: 50px auto; padding: 20px; }
        .progress-bar { background: #333; height: 20px; border-radius: 10px; margin:
20px 0; }

```



```
        setTimeout(loadQuestion, 1000);
    }
    loadQuestion();
</script>
</body>
</html>
```

- **Marking:** 1 mark for question display with options, 1 mark for score tracking and correct/incorrect highlighting, 1 mark for progress bar, 1 mark for final score with message, 1 mark for CSS styling with animations.

Part C: Notebook Maintenance (5 marks)

1. Neatness and organisation of notebook (2 marks)
2. Completion of all assigned classwork and homework (2 marks)
3. Proper headings, dates, and indexing (1 mark)